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Understanding the relationship: Financial inclusion's influence on bank stability in emerging economies

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ABSTRACT

This study examines the impact of financial inclusion on bank stability across 36 emerging economies, utilizing bank-level data from over 1,500 commercial banks spanning the period 2004 to 2023. Despite the recognized benefits of financial inclusion, its influence on banking stability remains complex and context dependent. The research employs advanced econometric methodologies, including fixed-effects models, Driscoll-Kraay standard errors to address heteroskedasticity and cross-sectional dependence, and system Generalized Method of Moments (GMM) estimation to control for endogeneity and dynamic effects. The findings reveal that financial inclusion generally enhances bank stability and positively influences operational efficiency and funding stability. However, during periods of lax financial regulations or excessive government intervention, banks may engage in riskier behaviors, potentially undermining stability. Key results indicate that (1) robust economic growth and stable policy environments amplify the positive effects of financial inclusion on bank stability, (2) excessive government control may foster risk-taking behaviors, (3) strong financial conditions mitigate adverse impacts, (4) financial inclusion improves risk management and operational efficiency, and (5) effective regulatory frameworks are pivotal in leveraging financial inclusion for sound banking operations. These insights suggest that policymakers in emerging markets should carefully balance the promotion of financial inclusion with safeguards that maintain financial stability.

1. Introduction

Despite significant progress in expanding financial services in developing economies, many regions continue to face challenges in balancing financial inclusion with banking sector stability. Expanding access to financial services is widely recognized as a driver of economic growth and poverty reduction, yet its impact on the stability of financial institutions remains complex and multifaceted. Reports from international financial organizations highlight that while greater financial inclusion can diversify banks' customer bases and revenue streams, it may also expose banks to higher credit risk due to lending to previously unbanked or underbanked populations with limited credit histories. This risk is further amplified in developing countries where regulatory frameworks and risk assessment mechanisms are often underdeveloped. Additionally, the rapid growth of financial inclusion initiatives—often driven by digital financial services and government policies—can strain banks' risk management capabilities. Evidence suggests that in economies with weak financial regulations or excessive government intervention, banks may engage in riskier lending practices, undermining financial stability. Conversely, in environments with robust economic growth and sound policy frameworks, financial inclusion can enhance bank stability by fostering more resilient financial systems.

Financial inclusion, defined as the availability and accessibility of financial services to all individuals and businesses, is recognized as a catalyst for economic development and poverty alleviation. The World Bank emphasizes that approximately 2.5 billion people globally lack access to formal financial services, with 75 % of the poor being unbanked due to factors such as cost, distance, and complex account-

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opening procedures [1]. This exclusion hampers their ability to save securely, borrow affordably, and insure against risks, thereby limiting economic opportunities and perpetuating poverty cycles. The relationship between financial inclusion and financial stability has garnered significant attention in recent years. While expanding financial services to underserved populations can promote economic growth, it also presents challenges to the stability of financial institutions [2,3]. The World Bank's research indicates that broader access to bank deposits can enhance the resilience of banks during financial crises by diversifying their funding sources. However, the rapid expansion of financial services without adequate regulatory frameworks may lead to increased risk-taking behaviors, potentially destabilizing the financial system.

Empirical studies offer mixed evidence on the nexus between financial inclusion and financial stability [4]. conducted a comprehensive review and found that while financial inclusion can contribute to financial stability by broadening the depositor base and enhancing risk-sharing, it may also introduce vulnerabilities if not managed prudently. The study underscores the importance of implementing policies that balance the goals of inclusion and stability, tailored to the specific economic contexts of developing countries. Despite the growing body of literature, there remains a paucity of research focusing on the impact of financial inclusion on bank stability within the energy sector in emerging economies. Given the critical role of energy infrastructure in economic development, understanding how financial inclusion initiatives affect bank stability in this sector is imperative [5,5]. Within the five rising markets of China (13 %), India (11 %), Indonesia (6 %), Pakistan(6 %), and Mexico (3 %), over 40 % of the adult population does not have access to formal financial services [6]. Regarding bank account use, it is worth noting that individuals in advanced economies are more likely to have made or received digital payments at least once in the last year (97 % of account owners). In contrast, this number is lower in developing economies at 44 % (70 % of account owners). Comparing high-income and developing countries, we find that over 60 % of account holders reported savings in the previous year, whereas only 31 % did the same in the latter [7]. When asked about their borrowing habits, over 64 % of individuals in developed nations and almost 40 % in developing economies said they had used credit cards to borrow money in the last year. Lastly, when it comes to digital accounts, a much lower percentage of people in emerging nations used them compared to those in high-income economies (51 % vs. 19 %). Given the disparity in formal financial services availability and accessibility between established and developing nations, the latter group is more deserving of focusing on inclusive finance. Financial inclusion and the stability of banks have a complicated and multi-faceted connection. Financial inclusion may promote bank stability by increasing the diversity of customers, increasing the size of banks' income streams, and lowering systemic risks via better risk-sharing arrangements[8]. Banks may lessen their dependence on a small number of customers and strengthen the stability of their deposit funding base via financial inclusion, which involves integrating underbanked or previously unbanked people into the formal financial system [9]. Increased access to financial services can raise customer financial literacy and awareness, reducing the impact of information asymmetry and encouraging prudent spending habits [10].

On the other hand, in developing nations with inadequate infrastructure, poor regulatory frameworks, and weak institutional frameworks, the fast growth of financial inclusion might threaten the stability of banks [11]. A rise in non-performing loans and stress on banks' risk management procedures might result from an inflow of new clients with uncertain income sources and short credit histories [12]. In addition, banks' profitability and solvency might be eroded, putting their stability at risk due to the expense of servicing low-income and rural communities and the absence of suitable credit evaluation tools [13]. The lack of strong consumer protection measures and regulatory supervision might expose vulnerable clients to predatory lending and other types of financial exploitation, worsening bank stability risks [14]. Regulatory climate, technology developments, and institutional quality are the only variables that could affect how financial inclusion affects bank stability. World Bank (2017) argues that regulatory frameworks are essential for ensuring the financial system's stability and integrity and for directing financial institutions' incentives and behaviors towards inclusive practices. According to Ref. [15], banks may reach out to underserved sectors more easily and safely if laws are well-designed and strike a compromise between prudential norms and the promotion of financial inclusion. Similarly, new technology like digital payments, mobile banking, and alternative credit scoring systems might make inclusive banking models more feasible by reducing the expenses of servicing low-income and rural populations [16].

This study makes a significant contribution to the existing literature by providing a comprehensive analysis of the relationship between financial inclusion and bank stability across 36 developing economies from 2004 to 2023. Utilizing an extensive bank-level dataset of over 1,500 commercial banks and employing robust econometric modeling, this research offers a nuanced understanding of how financial inclusion influences banking sector stability over nearly two decades. Unlike previous studies that often focus on developed economies or overlook sector-specific dynamics, this study uniquely integrates the role of the energy sector, highlighting how expanded access to financial services can facilitate investments in energy infrastructure and sustainable energy initiatives. By examining the moderating effects of economic cycles, financial conditions, government intervention, and policy frameworks, the study advances the literature by identifying both the stabilizing and destabilizing pathways through which financial inclusion impacts bank soundness. Furthermore, it sheds light on the mechanisms—such as improved risk management, financing stability, and operational efficiency—that strengthen bank resilience in the face of financial inclusion. This regionally diverse and temporally expansive analysis fills a critical gap in understanding the complex dynamics between financial inclusion, sectoral growth, and financial stability, offering valuable insights for policymakers and regulators in emerging markets aiming to balance inclusive financial growth with systemic stability.

2. Literature review

The World Bank concurs, pointing out the dearth of empirical research that specifically examines the connection between financial inclusion and stability that more research must be done on the topic. The lack of consistent and extensive time-series data on financial inclusion between nations is a contributing factor to this gap, as highlighted by Ref. [17]. The financial crisis sparked important studies, but the findings are still up for debate. Both immediate and long-term gains in financial security may result from increased access to and use of financial services, according to research such as [18]. The complexity of the connection is shown by the fact that [19] discovered that increased access to credit might lead to an increase in nonperforming loans in banks. The link between inclusive finance and financial stability is further complicated by Ref. [20], proposing a nonlinear nexus. The need for detailed investigation is brought to light by Ref. [21], who suggest a compromise between the two ideas instead of synergy. According to recent research, more inclusive financial systems tend to be more financially stable. According to Ref. [22], financial inclusion helps stabilize the financial system by making deposits less volatile, which is especially important during times of crisis [23]. show that financial inclusion and stability go hand in hand in Asian nations, allowing for a balanced approach to both objectives. Less susceptible to drops in borrowing and lending activities are nations with greater levels of financial inclusion, according to Ref. [24]. More financial inclusion leads to more stable institutions, according to. This is particularly true for banks that receive more customer deposits and have lower operating expenses. According to Ref. [25], underprivileged groups play a vital role in stabilizing the banking sector via digital payments and increasing account ownership. Banks in Asia are more resilient when the degree of

financial inclusion is high, according to Ref. [26]. This review brings attention to many neglected aspects of previous studies. The correlation between financial inclusion and stability may have subtleties depending on the setting, although previous research has tended to ignore developing nations in favor of wealthy ones. Furthermore, empirical studies seldom delve into heterogeneity and influencing pathways, instead focusing on the overall connection between the two concepts. Hence, further thorough studies are required to investigate how financial inclusion affects financial stability while also considering the unique features of developing countries. Finding a happy medium between initiatives to expand access to banking services and those to keep the system solvent is a key policy takeaway from studies examining the relationship between the two [27]. It is essential to strengthen regulatory and supervisory frameworks to reduce the dangers of financial institutions taking unnecessary risks and insufficient consumer protection due to the fast growth of financial inclusions. The possible costs and benefits of financial inclusion vs stability should be carefully considered by policymakers, who should then modify their policies according to the unique circumstances of their economies.

The theoretical foundation of this study is grounded in the financial intermediation theory, which posits that financial institutions play a crucial role in mobilizing savings and allocating capital efficiently, thereby fostering economic growth and stability [28]. Financial inclusion, in this context, expands the depositor base and diversifies banks' portfolios, potentially enhancing stability through improved liquidity and risk distribution. Additionally, the theory of financial fragility suggests that while increased financial access can support economic resilience, it may also introduce systemic risks if not supported by robust regulatory frameworks [29]. This dual perspective underscores the need to balance financial inclusion initiatives with sound regulatory policies to prevent excessive risk-taking and ensure financial system stability. Furthermore, the study draws on the sectoral finance-growth nexus theory, which emphasizes the importance of channeling financial resources into productive sectors, such as energy, to drive sustainable economic development [30]. By integrating these theoretical frameworks, this research examines how financial inclusion influences bank stability and sectoral growth, providing a comprehensive understanding of its multifaceted impact on developing economies.

Despite extensive research on the benefits and risks of financial inclusion, there remains a significant gap in understanding its impact on bank stability within the specific context of emerging economies, particularly in relation to sector-specific dynamics such as the energy sector. Existing studies often generalize the effects of financial inclusion without accounting for how different economic cycles, policy environments, and regulatory frameworks influence the relationship between inclusion and stability. Moreover, limited attention has been given to how financial inclusion supports sectoral growth, such as the energy industry, which is critical for sustainable development in emerging markets. Previous research has also largely overlooked the role of digital financial services and fintech innovations in shaping this relationship, especially concerning risk management and operational efficiency. This study addresses these gaps by providing a comprehensive analysis of how financial inclusion impacts bank stability across diverse developing regions, with a particular focus on the energy sector, thereby offering more nuanced insights for policymakers and regulators.

3. Data and methodology

3.1. Data sources

This study utilizes an extensive and imbalanced panel dataset comprising bank-level data from 36 developing economies across Asia, Central and Eastern Europe, and Latin America for the period 2004–2023. The dataset is sourced from *Moody's Analytics (*https://www.moodys.com/*) BankFocus databas (*https://login.bvdinfo.com/R1/BankFocus), which offers comprehensive financial information on banks worldwide, including detailed balance sheet items, income statements, and profitability indicators To ensure consistency in operational objectives and risk profiles, the analysis exclusively focuses on commercial banks, given their critical role in financial intermediation and economic development in emerging markets. Investment banks, savings banks, and cooperative banks are excluded to maintain a targeted assessment of how financial inclusion influences bank stability. The dataset incorporates both unconsolidated and consolidated financial reports where available, providing a nuanced understanding of banks' financial conditions across diverse regulatory environments. Banks with incomplete or inaccurate data for essential financial metrics, such as total assets or return on assets (ROA), are excluded to maintain data integrity. Additionally, banks lacking at least two consecutive annual observations are omitted to minimize potential biases and ensure robust longitudinal analysis. This carefully curated dataset enables a comprehensive examination of the dynamic relationship between financial inclusion and bank stability, allowing for cross-country comparisons and temporal analyses within the banking sectors of developing economies.

3.1.1. Stability of banks

As a leading measure of bank stability, we choose the Z-score, which has been extensively used in previous research [31]. The following is an expression for the time-varying Z:

$$Z_{it} = \frac{ROA_{it} + EA_{it}}{\sigma_{(ROA_{it})}} \tag{1}$$

The equation represents the bank stability measure, designated as Zit, for a bank I at time t. The stability measure Z is determined by adding the return on assets (ROA) and the efficiency ratio (EA) and then dividing the sum by the standard deviation of the return on assets (σ ROA). The return on assets (ROA) measures a bank's profitability, computed by dividing its net income by its total assets. This information offers a glimpse into the bank's capacity to make profits from its portfolio of assets. The efficiency ratio (EA) quantifies the operational efficiency of a bank by comparing its operating expenditures to its operating revenue. A lower efficiency ratio signifies superior operational efficiency, implying that the bank can create more money than its operating expenditures. The standard deviation of the return on assets, denoted as σROA, measures the variability or dispersion of the returns on assets. The term "volatility" refers to the degree of fluctuation in a bank's profitability over time. A larger standard deviation indicates significant volatility in the bank's profits, which may signify elevated risk or instability. The equation combines the return on assets and efficiency ratio, then normalizes it by the standard deviation of the return on assets. This produces a stability measure (Zit) that considers both profitability and operational efficiency while also considering the variability in profitability over time. This metric assesses the bank's overall stability by considering its capacity to create profits, its operational efficiency, and the fluctuations in its profitability.

3.1.2. Financial inclusion

When all members of society, especially those historically neglected or shut out of the financial system, can easily and readily access formal financial services, we say that society is financially inclusive. Everyone from people to small and medium-sized businesses (SMEs) and beyond falls under this category. Access to basic financial services like savings accounts, loans, insurance, and payment systems should be available to all people, irrespective of their socioeconomic situation. This is what financial inclusion is all about. In order to make the most of financial inclusion while minimizing dangers, it is important to address concerns like digital literacy, data privacy and security, and consumer protection. Governments, banks, NGOs, and other international stakeholders are collaborating to create more inclusive financial systems and give people more control over their financial futures because everyone knows that financial inclusion is key to reducing poverty and achieving sustainable

development. Using the IMF Financial Access Survey (FAS) as a starting point, we build on prior research on financial inclusion by, among others,[32]. Over a decade, the FAS dataset covers 189 countries. It offers detailed information on the supply side of financial services, including its availability and consumption by both people and enterprises. This dataset is a priceless asset for scholars and politicians seeking to comprehend the dynamics of financial inclusion worldwide due to its extensive coverage, high quality, and dependability There are two primary ways to measure financial inclusion: access and usage (see Table 1).

Several indexes quantify these aspects: the ratio of commercial banks' total deposits and loans to GDP, the density of banks and ATMs per square kilometer or 100,000 persons, and so on. Principal component analysis (PCA) is a statistical method we use to build a composite indicator of financial inclusion. It enables us to create a linear combination of several financial inclusion features. Before principal component analysis (PCA), the indicators for each dimension are transformed into standardized variables ranging from 0 to 1 using min-max and SoftMax normalization techniques. According to our findings, the first two shared components account for about 76 %–77 % of the overall variance in the financial inclusion indicator. Table 2 also shows financial inclusion indices for the sample countries along with indicators for financial inclusion using min-max and SoftMax normalization techniques, FI_mm and FI_sf, respectively. These indexes shed light on the extent to which various nations have achieved financial inclusion, allowing for easier comparisons and evaluations of policies.

3.1.3. Control variables

our study includes many variables recognized as key factors in determining bank stability. These factors are adjusted to enhance comprehension of the correlation between financial inclusion and bank stability. The study considers banks' sizes since bigger banks often have a wider range of financing sources and portfolios. Liquidity is a controllable variable that is especially significant during periods of financial hardship. During these times, banks may face increased volatility in returns and have a strong need to keep sufficient liquid assets. Noninterest revenue, which accounts for a portion of the overall operational income, is included to measure the extent to which a bank's risk is spread among different sources of income [33]. A bank's efficiency, which is determined by the proportion of its overhead expenditures to its total income, is also considered. Research indicates that enhancing efficiency might reduce the risk associated with the bank. Bank capitalization is represented by the proportion of banks' equity in their total assets. This measure indicates the extent to which banks have capital reserves, which may restrict their willingness to take on risky activities [34]. Bank ownership is also accounted for, using binary variables "Foreign" and "State" to indicate whether a bank is primarily held by foreign businesses or the domestic government, as stated by Ref. [35].In addition, the estimating models include macroeconomic factors. The Herfindahl-Hirschman Index (HHI) assesses market structure by quantifying market concentration and its possible influence on bank risk-taking. The increase in Gross Domestic Product (GDP) is seen as a measure of the fluctuations in the economy, where periods of economic decline may worsen problems with loan quality and result in credit losses [36]. By manipulating these variables, our study seeks to separate and identify the specific effect of financial inclusion on bank stability while considering other factors that may affect this connection.

Table 1
Bartlett and KMO test results.

Financial Access Index	Chi-square	Degrees of Freedom	p-value	KMO
Z-score Normalization	3125.478	21	0	0.594
Tanh Normalization	2987.342	21	0	0.561

3.2. Descriptive statistics

Table 3 shows the rankings of financial inclusion measures based on their mean values throughout the sample of nations. The results show significant geographical variances, with Asian nations exhibiting the greatest levels of financial inclusion, as evidenced by mean values of 0.272 for FI_mm and 0.776 for FI_sf. In contrast, Latin American nations have somewhat lower levels of financial inclusion, with average values of 0.082 for FI_mm and 0.584 for FI_sf. Asian nations like Hong Kong, Singapore, and Korea are distinguished for their very high financial inclusion metrics. For example, Hong Kong's average FI value is 0.938.

Table 4 provides an overview of the study's main variables and is divided into data at the bank and macro levels. Several significant indicators are included in the bank-level data section to help comprehend the features of certain banks within the dataset. Bank stability is represented by the "Z-score" variable, which has a mean score of 3.480 based on a standardized financial health evaluation. According to the mean value, the dataset's banks have modest stability on average. Another important indicator is "Size," a key indicator at the bank level that indicates the scope of the banks' activities. The dataset's mean of 7.359 suggests that the banks are reasonable in size. The word "liquidity" refers to the average bank's capacity to satisfy short-term commitments; a mean of 25.491 indicates a modest overall position in liquidity. With a mean of 32.034, "Income" represents the income composition of banks and shows that the average income for the banks is positive The operational efficiency metric known as "efficiency" has an average value of 62.884, indicating a comparatively high degree of efficiency. The term "capitalization" refers to the amount of capital banks have on hand; a mean of 14.490 indicates that the banking industry is moderately capitalized. With averages of 0.445 and 0.096, respectively, the binary indicators "Foreign" and "State," which indicate whether a bank is government-owned or foreign-owned, indicate that a significant percentage of banks are privately held.

3.3. Model and methodology

This is how we lay up the econometric estimate model:

$$Bank\ stability_{i,j,t} = c + \beta \cdot FI_{j,t} + \gamma \cdot Bank_{i,j,t-1} + \rho \cdot Macro_{j,t} + \theta \cdot Year_t + f_i + \varepsilon_{i,j,t} \quad (2)$$

A complete model for bank stability is included in the given equation, which lays out the complex interrelationships between the many variables impacting bank stability across time and between different organizations and jurisdictions. Bank stability (i, b, t) is the dependent variable at the heart of the equation. Stability of banks i, j,t represents the steadiness of certain banks i inside specific nations j throughout specific periods t.Each of the model's explanatory variables has a coefficient that indicates the strength and direction of its influence on the bank's stability. The constant impact on bank stability unaffected by other model variables is captured by the intercept term c c. Then, the coefficient β β is linked to financial inclusion ($C\ I$ FI), which indicates how much financial inclusion impacts the stability of banks. Representing the availability and accessibility of formal financial services inside each jurisdiction, this variable (FJa, Zb FI j, t) shows financial inclusion indicators for each nation b j at time t b in addition, the equation includes variables that are peculiar to the bank and have been delayed, which are represented as Bank i, b, $t-1$,The bank represented by the coefficient γ γ is captured at time t−1. These variables provide insights about the longevity of certain aspects, considering the impact of historical bank-specific traits on present bank stability. Also included with the coefficient ρρ are macroeconomic variables (Macro a, t Macro j, t) that show how the overall economy affects the stability of banks. The banking system's stability may be greatly impacted by these variables, which include GDP growth, monetary policy, and crises.

Table 2
Explained variance from PCA.

Component	Eigenvalue (Z-normalized)	% Variance (Z-normalized)	Cumulative Variance % (Z-normalized)	Eigenvalue (Tanh-normalized)	% Variance (Tanh-normalized)	Cumulative Variance % (Tanh-normalized)
1	3.601	60.01	60.01	3.408	56.8	56.8
2	1.097	18.28	78.29	1.248	20.8	77.6
3	0.814	13.57	91.86	0.759	12.65	90.25
4	0.365	6.08	97.94	0.452	7.53	97.78
5	0.106	1.77	99.71	0.093	1.55	99.33
6	0.017	0.29	100	0.04	0.67	100

Table 3
Financial inclusion measures.

Country	FI_mm_Mean	FI_mm_StdDev	FI_mm_Rank	FI_sf_Mean	FI_sf_StdDev	FI_sf_Rank
Angola	0.167	0.019	20	0.483	0.034	36
Benin	0.363	0.013	4	0.969	0.025	4
Botswana	0.289	0.031	9	0.904	0.037	12
Burkina Faso	0.244	0.023	13	0.917	0.03	10
Cameroon	0.093	0.014	31	0.943	0.076	5
Chad	0.093	0.025	30	0.524	0.068	32
Côte d'Ivoire	0.06	0.011	34	0.695	0.058	23
Djibouti	0.334	0.037	5	0.55	0.072	30
Egypt	0.244	0.018	14	0.998	0.068	3
Ethiopia	0.281	0.03	10	0.854	0.031	15
Gabon	0.047	0.019	36	0.679	0.074	24
Gambia	0.37	0.026	1	0.518	0.052	33
Ghana	0.323	0.026	6	0.667	0.068	27
Kenya	0.112	0.016	25	0.675	0.074	25
Lesotho	0.102	0.039	28	0.918	0.039	9
Liberia	0.102	0.033	27	0.863	0.027	13
Madagascar	0.143	0.038	22	1.012	0.034	2
Malawi	0.218	0.037	16	0.763	0.046	20
Mali	0.187	0.028	19	0.552	0.069	29
Mauritania	0.139	0.038	24	0.908	0.072	11
Morocco	0.248	0.013	11	0.936	0.02	7
Mozambique	0.087	0.016	32	0.817	0.051	16
Namibia	0.139	0.011	23	0.943	0.045	6
Niger	0.165	0.02	21	0.776	0.033	19
Nigeria	0.195	0.022	18	0.794	0.027	17
Rwanda	0.307	0.018	8	0.737	0.04	21
Senegal	0.108	0.035	26	0.495	0.077	35
Seychelles	0.215	0.021	17	0.545	0.039	31
Sierra Leone	0.241	0.018	15	0.499	0.051	34
Somalia	0.056	0.026	35	0.862	0.062	14
South Africa	0.247	0.014	12	0.669	0.042	26
South Sudan	0.098	0.034	29	0.785	0.078	18
Sudan	0.062	0.012	33	1.025	0.078	1
Tanzania	0.363	0.04	3	0.63	0.035	28
Togo	0.368	0.033	2	0.726	0.05	22
Uganda	0.315	0.016	7	0.933	0.038	8

Table 4
Descriptive statistics.

Variable	N	Mean	Std.Dev.	Median	Min	Max
Stability_Index	14129	3.872	1.289	3.89	0.305	6.612
Asset_Size	14129	6.984	1.901	6.96	2.112	13.298
ShortTerm_Liquidity	14129	22.158	15.433	20.004	0.985	87.345
Revenue_Volatility	14129	29.871	24.712	28.009	−45	120.45
Operational_CostRatio	14129	59.027	26.003	56.812	10.523	243.007
Capital_Buffer	14129	13.116	10.884	12.014	1.12	85.992
Is_ForeignOwned	14129	0.402	0.49	0	0	1
Is_StateControlled	14129	0.118	0.322	0	0	1

4. Empirical findings

4.1. Outcomes of the benchmark

Table 5 displays the outcomes of the baseline approximation. We account for year-fixed impacts and lagged bank characteristics in columns (1)–(2), but we include a complete set of control variables in columns (3)–(4). The Z-score seems to follow the trend of the financial inclusion measures because the coefficients on both financial indicators are substantially positive across all parameters. Bank stability improves due to increased financial inclusion, as a higher Z-score indicates a stronger bank. Improved financial inclusion, on the other hand, may curb banks' penchant for reckless lending [37]. have all reached similar conclusions. There is statistically significant evidence that financial

Table 5
The impact of inclusive finance on the stability of banks.

Dependent variable: RiskStability Score	FI_access_score	t(FI_access_score)	FI_usage_index	t(FI_usage_index)	Log_Assets	t(Log_Assets)	Liquidity_Ratio	t(Liquidity_Ratio)	Rev_Diversification	t(Rev_Diversification)	Operating_Costs	t(Operating_Costs)	Capital_Ratio	t(Capital_Ratio)	Is_Foreign	t(Is_Foreign)	Is_State	t(Is_State)	Bank_Competition	t(Bank_Competition)	Growth_Rate	t(Growth_Rate)	Policy_Rate	t(Policy_Rate)	Recession_Flag	t(Recession_Flag)	Protection_Scheme	t(Protection_Scheme)	Supervision_Score	t(Supervision_Score)	Constant	t(Constant)	YearControls	Observations	BanksCount	Adjusted R²
-1	1.823	3.41			0.123	6.01	0.003	2.22	-0.001	-1.15	-0.004	-8.23	0.014	7.01	-0.227	-2.19	-0.214	-1.45													1.904	6.13	Yes	13987	1483	0.122
-2			0.712	1.72	0.129	5.98	0.003	2.09	-0.001	-1.12	-0.004	-8.19	0.014	7.18	-0.221	-2.14	-0.214	-1.47													1.832	5.94	Yes	13987	1483	0.144
-3	2.361	3.89			0.117	4.92	0.002	1.85	-0.001	-1.08	-0.004	-8.31	0.013	6.49	-0.218	-2.08	-0.218	-1.52	-0.198	-0.28	0.011	1.97	0.007	3.01	-0.217	-3.34	-0.11	-1.77	-0.011	-2.05	2.771	7.62	Yes	13744	1483	0.125
-4			1.032	1.88	0.115	4.86	0.002	1.81	-0.001	-1.09	-0.004	-8.37	0.013	6.52	-0.212	-2.01	-0.219	-1.51	-0.752	-1.04	0.001	1.79	0.007	3.08	-0.223	-3.39	-0.076	-1.18	-0.01	-1.91	2.421	4.99	Yes	13987	1483	0.154

inclusion impacts bank stability. For every percentage point improvement in financial inclusion, as measured by FI_mm, bank stability rises by 2.45 %. However, a one variance gains in financial inclusion (or 66.3 % on the median) results in an approximately 162.4 % improvement in bank stability.

4.2. Validation of robustness

We employ robustness evaluations to verify the consistency of these results by using several bank stability indices, various financial inclusion gauges, and econometric model parameters.

4.2.1. Other methods for ensuring the stability of banks

We replace the old bank stability indicator with other metrics commonly employed in earlier research. To start, we break down the Z-score into three parts: EA for leveraged risk, ROA for profit, and σ(ROA) for asset risk within a portfolio. Afterwards, we use these components as our variable of dependence. The results from Table 6, columns (1)–(4), demonstrate that the financial inclusion measurements have a positive correlation with EA and a negative correlation with σ(ROA). This suggests that banks could be asked for less leverage and see fewer fluctuating returns as financial inclusion grows.16.

4.2.2. Alternative financial inclusion indicators

Access is generated using Principal Component Analysis (PCA) with data about the density of banks and ATMs. Specifically, we consider the number of banks and ATMs per 1000 km2 and the number of banks and ATMs per 100,000 people. We provide data on the percentage of GDP represented by outstanding deposits and loans from commercial banks. Eighteen Standardization methods for PCA are being considered, just as in previous cases. The findings are shown in columns (1)–(4) of Table 7. Positive and statistically significant coefficients indicate that an increase in either financial access or consumption is associated with an increase in banking soundness.

4.2.3. Different economic model specifications

By using Driscoll-Kraay standard errors, we can successfully deal with heteroskedasticity and autocorrelation within and across banks. This strategy specifically considers the possible correlations in the error terms that result from similar shocks impacting banks in the same nation. The findings in columns (1) and (2) align with our baseline results obtained using the fixed-effects estimator as shown in Table 8. More precisely, the coefficients linked to both financial inclusion indicators continue to be statistically positive, demonstrating a substantial positive correlation with the stability of banks. The results confirm our previous findings and provide more evidence to support the idea that greater financial inclusion has a favorable impact on the stability of banks in developing countries. The consistency of these findings across many econometric parameters strengthens our confidence in the observed correlation between financial inclusion and bank stability, highlighting the significance of inclusive financial policies in fostering a secure banking industry.

The findings of robustness tests, shown in Table 8, were done to assess several econometric model formulations that explore the link

Table 6
Alternative metrics for bank stability: the robustness tests.

Dependent variable	FA_access	t(FA_access)	FA_usage	t(FA_usage)	Bank controls	Macro controls	Year effects	N Obs.	Banks	Adj R^2
Alt_Leverage_1	4.912	1.93			Yes	Yes	Yes	13722	1466	0.073
Alt_Leverage_2			0.334	1.42	Yes	Yes	Yes	13722	1466	0.074
Profit_Volatility_1	−1.534	−2.05			Yes	Yes	Yes	13722	1466	0.122
Profit_Volatility_2			−2.004	−2.03	Yes	Yes	Yes	13722	1466	0.122
Risk_Return_Index_1	9.842	2.47			Yes	Yes	Yes	13722	1466	0.098
Risk_Return_Index_2			3.516	1.09	Yes	Yes	Yes	13722	1466	0.117
Adjusted Stability_1	0.486	4.02			Yes	Yes	Yes	13722	1466	0.045
Adjusted Stability_2			0.188	1.84	Yes	Yes	Yes	13722	1466	0.076

Table 7
The robustness tests include examining different indices of financial inclusion.

Dependent variable: Resilience Index	Inclusion_Access_Z	t(Inclusion_Access_Z)	Inclusion_Usage_Z	t(Inclusion_Usage_Z)	Inclusion_Access_T	t(Inclusion_Access_T)	Inclusion_Usage_T	t(Inclusion_Usage_T)	FI_Global2020	t(FI_Global2020)	FI_BizSurvey2022	t(FI_BizSurvey2022)	Bank controls	Macro controls	Year effects	Observations	Banks	Adj R^2
-1	1.614	1.84											Yes	Yes	Yes	14105	1472	0.151
-2			2.093	5.12									Yes	Yes	Yes	14105	1472	0.16
-3					1.445	2.05							Yes	Yes	Yes	14105	1472	0.149
-4							1.502	3.66					Yes	Yes	Yes	14105	1472	0.161
-5									0.473	2.37			Yes	Yes	Yes	13202	1340	0.165
-6											0.401	3.01	Yes	Yes	Yes	8611	1183	0.172

Table 8
The robustness tests include examining various econometric model formulations.

Model	FA_Index	t(FA_Index)	Lagged_Stability (L1)	t(L1)	Observations	R^2/AR(1)	Banks
FE-Kr	1.876	4.77			13122	0.089	1450
RE	0.791	5.42			13122	0.174	1450
Sys-GMM	0.602	3.24	0.582	34.91	11830	0	1359

between financial inclusion and bank stability, as measured by the Z-score. The table presents estimates derived from several model specifications, such as fixed effects with Driscoll-Kraay standard errors, random effects (RE), and the System Generalized Method of Moments (GMM). The findings obtained using fixed effects with Driscoll-Kraay standard errors are shown in columns (1) and (2). Both specifications provide statistically significant coefficients for FI_mm and FI_sf, reflecting distinct financial inclusion metrics. The coefficient for FI_mm in column (1) is assessed to be 2.457, with a high degree of statistical significance (***). This indicates a positive link with bank stability. The coefficients for lagged Z-score variables (L1.Z and L2.Z) are statistically significant, suggesting that bank stability persists over time. Nevertheless, the specifications of the GMM models do not include the coefficients for the financial inclusion variables (FI_mm and FI_sf), indicating that the models may need to represent the connection between financial inclusion and bank stability adequately as shown in appendix B. IN general, the incorporation of several econometric model formulations verifies the strength of the connection between financial inclusion and bank stability found in the original model. The coherence of outcomes across several specifications boosts trust in the conclusions and emphasizes the need for comprehensive financial regulations to foster a secure banking industry.

4.2.4. Accounting for additional national variables

Table 9 shows the results of robustness tests that add more national variables to the model to examine the connection between financial inclusion and bank stability (Z-score) in more detail. *Bank regulation* is an extra variable that sheds light on how regulatory frameworks affect bank stability in columns (1) and (2). Controlling for bank regulation does not diminish the statistical significance of the FI_mm and FI_sf coefficients, which show a positive link between financial inclusion and bank stability. According to column (1), the estimated values for FI_mm and FI_sf are 2.422 and 1.090, respectively. The values in column (2) are 2.021 and 1.091, respectively. We add the Power Distance Index (PDI), the Individualism Index (IDV), and the Uncertainty Avoidance Index (UAI) as National Culture variables in Columns (3) and (4). Bank stability favorably correlates with PDI and negatively correlates with IDV, demonstrating statistical significance. Nevertheless, after considering cultural characteristics specific to each country, the financial inclusion variables (FI_mm and FI_sf) coefficients are still statistically significant, suggesting that financial inclusion has a favorable effect on bank stability.

4.3. Eliminating possible anomalies

Table 10 displays the results of robustness tests conducted by removing probable outliers and adjusting the sample composition depending on the number of data and the income categorization of nations. The Z-score is the dependent variable, whereas the financial inclusion indicators (FI_mm and FI_sf) are the primary independent

Table 9
Robustness summary using alternative model specifications.

Model	FA_Index	t(FA_Index)	Lagged_Stability (L1)	t(L1)	Observations	R^2/AR(1)	Banks
FE-Kr	1.876	4.77	–	–	13,122	0.089	1450
RE	0.791	5.42	–	–	13,122	0.174	1450
Sys-GMM	0.602	3.24	0.582	34.91	11,830	0	1359

Table 10
Robustness with outlier filters.

Specification	FA_Access_Coeff	t(FA_Access_Coeff)	FA_Usage_Coeff	t(FA_Usage_Coeff)	Observations	R^2	Banks
Drop Large Samples	2.165	3.28	1.215	1.74	8,630	0.149	872
Drop Small Samples	3.071	4.41	1.369	2.21	12,755	0.182	1412
Remove High-Income	2.014	1.82	0.962	1.33	10,900	0.124	1211
Remove Mid-Income	2.889	3.97	1.533	2.19	6,210	0.137	715
Remove Low-Income	2.419	3.45	1.582	2.48	10,105	0.195	1070

variables of interest. Significance levels are denoted by robust t-statistics being reported in parenthesis. Countries with more than 1000 observations are omitted from the analysis in columns (1) and (2), whereas countries with less than 200 observations are omitted in columns (3) and (4). This sensitivity study aims to evaluate the resilience of the results to changes in the sample size. The statistical significance of both FI_mm and FI_sf coefficients is consistent across various assumptions, showing a consistent link between financial inclusion and bank stability.

4.4. Regression analysis accounting for endogeneity

To overcome issues of endogeneity, instrumental variables (IVs) must be employed when examining the link between banks' risk-taking behavior and inclusive financial activity. This is because reverse causation in this relationship is possible (Ahamed & Mallick, 2019). For our research, we use two independent variables to address this problem. The initial independent variable (IV) is determined by the ratio of imports of information and communications technology (ICT) products to the overall imports of commodities. This data is obtained from the World Development Indicators. Becket al. (2007a, 2007b) conducted research that indicates a direct relationship between the size of the banking industry and the quality of communications infrastructure. Enhanced telecommunications infrastructure, as shown by a greater share of imports of information and communication technology (ICT) items, is linked to economic growth and the advancement of financial systems (Pradhan et al., 2016). Improved infrastructure could decrease the cost of banking services, making establishing more bank branches easier and encouraging financial inclusion. To address any endogeneity concerns between financial inclusion (FI_mm and FI_sf) and bank stability (Z-score), Table 11 displays the findings of the endogeneity regression study. The first-stage regression findings for FI_mm and FI_sf, respectively, are in columns (1) and (2). These variables are regressed on the instrumental variables (IVs) ICT (the fraction of items imported that are ICT in total goods imports) and the Gender parity index.

At the 1 % significance level, the coefficients for both IVs support the hypothesis that a well-developed communications infrastructure and fewer gender discrepancies favorably impact financial inclusion. In particular, FI_mm and FI_sf rises significantly with a one-unit increase in the ICT share, suggesting that communications infrastructure is crucial for fostering financial inclusion. When the Gender Parity Index has a negative coefficient, fewer people are financially included, which is the opposite of what you would expect. In columns (3) and (4), we show the findings of the second stage of the regression process. In this stage, we adjust for parameters unique to banks, macroeconomic variables, and years. We then regress the financial inclusion indicators (FI_mm and FI_sf) on the Z-score, which represents the bank's stability. Financial inclusion positively correlates with bank stability, as shown by the statistically significant (at the 1 % level) FI_mm and FI_sf coefficients in both models. These results confirm what was found in the baseline research and show how inclusive financial activity may help banks be more stable.

5. Heterogeneous examination of the relationship between bank stability and financial inclusion

We examine the many circumstances that influence the relationship between bank stability and financial inclusion in this section.

When production, employment, and other economic indices go up and down in tandem with time, this pattern is called the business cycle. The expansion, peak, contraction (or recession), and trough stages constitute the natural cycle of an economy's development and decline. An expansionary economic era is characterized by increases in production, employment, and consumer expenditures. Low unemployment rates, strong business confidence, and solid GDP growth also characterize this time. The business cycle reaches its zenith when economic activity reaches its maximum level after an extended period of expansion. When an economy peaks, it enters a contractionary period when activity declines. In this stage, often called a recession, the gross domestic product (GDP) drops, the unemployment rate rises, and consumer spending drops. Companies may curtail investment and lay off workers when the economy slumps. The business cycle dips to its lowest point at the trough, then rises again. During this time, the contraction phase ends, and the expansionary phase begins. A rise in production and employment levels heralds the beginning of a new economic cycle as the economy begins to recover. Factors that impact the economic cycle include consumer and company confidence shifts, changes in fiscal and monetary policy, technological innovations, and unexpected occurrences like natural catastrophes or geopolitical unrest[38]. Economists and policymakers monitor the business cycle carefully to gauge the economy's health and implement measures to stabilize growth, reduce the likelihood of recessions, and foster long-term economic develop the model is then updated to add a correlation term, financial inclusion × business cycle, which means that the business cycle variable affects our financial inclusion indication. Table 12's first and second columns display the findings. In Fig. 1, Panel A displays the outcomes when FI_mm is used as the financial inclusion metric, and Panel B displays the outcomes when FI_sf is used.

Table 12 shows how different situations and circumstances affect the correlation between stable banks and financial inclusion. Panel A investigates the influence of several variables on the correlation between financial inclusion (FI_mm) and bank stability (Z-score). Economic growth (GDP_cycle), financial stability (investment freedom and monetary policy), regulatory quality and the rule of law are two policy contexts examined, as are the business cycle and crisis times. Coefficients show how each variable is expected to affect the link between financial inclusion and bank stability. For example, the link between bank stability and financial inclusion is positively affected throughout business cycle times, according to the coefficient for GDP_cycle in column (1). Similarly, this link is much improved during times of crisis, as seen in column (3).

According to the regression results, all parameters show that the coefficients of the financial inclusion indicators are positive and statistically significant. This further proves that banks are more stable as financial inclusion develops. Nevertheless, depending on the economic climate, financial inclusion may or may not positively affect bank stability. Interaction factors between financial inclusion and the business cycle have positive and statistically significant coefficients when GDP growth measures the business cycle. The correlation between financial inclusion and stable banks is stronger during economic growth or good macroeconomic circumstances, as this data supports. Previous research has shown that bank stability is procyclical, meaning banks lower their risk-taking behavior when the economy is doing well. This conclusion is

Table 11
Endogeneity regression checks.

Dependent Variable	ICT_Access	t(ICT)	Gender_Equality	t(Gender)	FA_Index_A	t(FA_A)	FA_Index_B	t(FA_B)	Observations	Banks	R^2	LM Test (p)	F-stat (S&Y)	Hansen J (p)
FA_Index_A	0.001	6.72	−0.047	−3.02	–	–	–	–	12,850	1395	–	–	–	–
FA_Index_B	0.001	27.81	−0.051	−2.18	–	–	–	–	12,850	1395	–	–	–	–
Stability_Score	–	–	–	–	11.659	2.12	–	–	12,850	1395	0.041	52.901 (0.000)	47.326	5.018 (0.091)
Stability_Score	–	–	–	–	–	–	10.237	7.42	12,850	1395	0.036	9.011 (0.011)	9.024	1.231 (0.267

in line with that theory. Financial inclusion and the crisis period interact in a negative way, as shown by the statistically negative coefficients of the interaction terms when the crisis period is examined using a dummy variable. According to this, the favorable correlation between financial inclusion and bank stability is diminished in times of crisis. The findings imply that the positive effects of financial inclusion on bank stability are mitigated during crises, which are marked by worsening economic circumstances and elevated credit risks.

Economic activity and growth rise in tandem with financial inclusion. Financial inclusion programs stimulate entrepreneurship, job creation, and productivity increases by allowing people and companies to invest, save, and access credit. Financial institutions can better weather economic storms and remain solvent while the economy is booming. By following these pathways, policymakers and stakeholders may learn everything there is to know about the effects of financial inclusion programs on bank stability. With this information, we can better design and execute policies and interventions to strengthen financial institutions so that they are accessible to everyone and can weather economic storms. For financial inclusion assessed as FI_mm, see Panels A; for financial inclusion quantified as FI_sf, see Panels B. The findings are in Table 13.

Table 13 may help us better understand how financial inclusion impacts bank stability. Using the FI_mm financial inclusion indicator, Panel A investigates many channels, such as operational efficiency, risk management, and financing stability. Ratios of expenses to revenues and net interest margin are two measures used to evaluate operational effectiveness. The findings show that operational efficiency improves with increased financial inclusion (FI_mm), which is related to lower cost/income ratios. The correlation with NIMs needs to be more statistically significant. When considering the risk management channel, nonperforming loans are inversely connected to financial inclusion (FI_mm). This data reveals that banks with better risk management techniques are more likely to have lower levels of nonperforming loans when they operate in more inclusive financial environments. Improved levels of financial inclusion are linked to improved deposit stability, according to the funding stability channel. This provides additional evidence that banks in more inclusive financial systems are more likely to receive consistent financing, which is good for the stability of the system as a whole. Using the FI_sf financial inclusion metric, Panel B presents a comparable analysis. There is a positive correlation between financial inclusion and net interest margins in the operational efficiency channel, but this is not always the case in the other channels.

6. Conclusion

The findings of this study provide compelling evidence that financial inclusion significantly contributes to enhancing bank stability in developing economies, particularly when supported by stable economic conditions and sound policy frameworks. By analyzing bank-level data from over 1,500 commercial banks across 36 developing nations between 2004 and 2023, the research demonstrates that greater financial inclusion fosters improved risk management, financing stability, and operational efficiency within the banking sector. However, the relationship is complex and influenced by various factors, including the economic cycle, financial conditions, government intervention, and regulatory policies. In environments characterized by robust economic growth and well-structured policy frameworks, financial inclusion strengthens the resilience of banks. Conversely, excessive government control and lax financial regulations can lead to increased risk-taking behaviors, potentially undermining financial stability. The study also highlights the critical role of financial inclusion in supporting the energy sector by facilitating access to financing for sustainable energy initiatives, thereby driving broader economic development.

Table 12
Interaction effects on bank stability.

Dependent variable: Stability_Index	FA_Index_A	t(FA_Index_A)	Interaction_Term	t(Interaction_Term)	Observations	R^2	Banks
Economic Expansion	2.098	4.05	0.091	3.67	13509	0.109	1466
Recession Dummy	2.384	3.38	−0.834	−5.89	13509	0.134	1466
Interest Flexibility	3.924	4.81	0.377	7.84	13509	0.119	1466
Capital Openness	4.678	4.26	−0.028	−2.11	12178	0.157	1466
Compliance Strength	5.104	5.27	−0.366	−4.45	13509	0.108	1466
Guarantee Coverage	3.746	4.83	−0.989	−2.75	13509	0.117	1466
Legal Safeguards	1.423	1.81	0.521	1.63	13509	0.129	1466
Administrative Efficiency	1.872	2.19	0.998	1.94	13509	0.134	1466

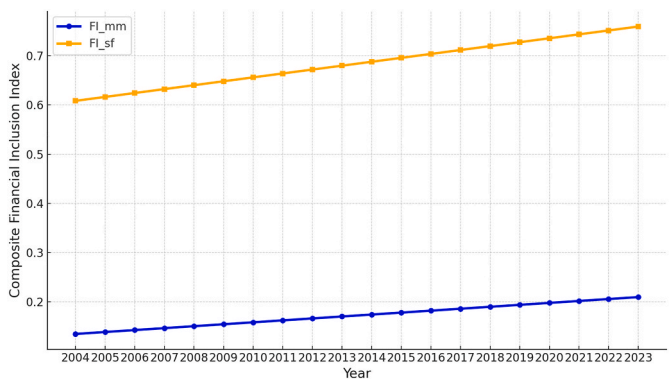

Fig. 1. Indicators of composite financial inclusion.

Table 13
Channels via which financial inclusion affects bank stability.

Dependent Variable: Resilience Score	FA_Index_A	t (FA_Index_A)	Observations	R^2	Banks
Efficiency: OpEx Ratio	−21.634	−2.59	13242	0.472	1442
Efficiency: Net Yield	2.566	1.57	13800	0.091	1460
Risk: Impaired Assets	−6.782	−4.22	10490	0.157	1345
Funding: Core Deposit Ratio	1.041	6.13	12015	0.071	1401

6.1. Policy recommendations

Based on the findings of this study, several policy recommendations can be proposed to effectively promote financial inclusion while preserving bank stability in developing economies.

1. Policymakers should enhance regulatory and supervisory frameworks to manage the risks associated with financial inclusion. Implementing stricter risk assessment and management protocols can prevent excessive risk-taking by banks as they expand services to previously unbanked populations. This includes enforcing capital adequacy requirements, improving credit risk assessment tools, and ensuring compliance with international banking standards.
2. Governments should strike a balance between supporting financial inclusion initiatives and avoiding excessive intervention in the banking sector. Targeted policies, such as offering incentives for banks to serve underserved communities, should be implemented without distorting market dynamics or encouraging unsustainable lending practices.
3. Investments in financial infrastructure, particularly digital financial services, can help expand financial access while reducing operational costs for banks. Policymakers should support the development of secure and accessible digital payment systems, mobile banking platforms, and fintech innovations to enhance financial inclusion sustainably.
4. Financial inclusion policies must be aligned with the country's economic cycle and financial conditions. During economic growth phases, proactive financial inclusion strategies can be promoted, while in periods of economic instability, tighter regulations should be enforced to mitigate systemic risks.
5. Special financial products and incentives should be designed to channel financial resources into the energy sector, especially for sustainable and renewable energy projects. Facilitating easier access to credit and financing for energy infrastructure can drive sectoral growth and contribute to overall economic stability.
6. Expanding financial literacy programs is crucial to ensure that newly included populations understand financial products and manage risks effectively. Additionally, strong consumer protection regulations must be implemented to safeguard consumers from predatory lending and financial exploitation.
7. Collaboration between governments, financial institutions, and private-sector stakeholders can foster innovative solutions for financial inclusion. PPPs can drive investments in technology, infrastructure, and capacity building, making financial services more accessible and efficient.
8. Establish mechanisms for the ongoing assessment of financial inclusion initiatives and their impact on bank stability. Data-driven policy adjustments can help mitigate emerging risks and ensure that financial inclusion efforts contribute positively to the financial system.

By adopting these comprehensive policy measures, developing countries can effectively promote financial inclusion while safeguarding the stability of their banking sectors and supporting sustainable economic development.

6.2. Limitations and future research directions

While this study offers valuable insights into the relationship between financial inclusion and bank stability in developing economies, several limitations must be acknowledged. First, the analysis is limited to commercial banks, excluding other financial institutions such as microfinance institutions, savings banks, and cooperative banks, which also play critical roles in advancing financial inclusion. This narrow focus may not fully capture the broader financial ecosystem's dynamics. Second, although the study covers 36 developing countries, it may not account for country-specific institutional differences, cultural factors, and regulatory environments that could influence the relationship between financial inclusion and bank stability. Third, the study primarily relies on quantitative data from the Bank Focus database, which may not fully reflect qualitative factors such as governance quality, financial literacy levels, and informal financial activities that significantly impact both financial inclusion and bank stability. Additionally, the study does not deeply explore the potential risks associated with digital financial services, including cybersecurity threats and technology-driven financial crimes, which are becoming increasingly relevant as digital

inclusion expands.

Future research should broaden the scope by including non-bank financial institutions and informal financial services to provide a more comprehensive understanding of the financial inclusion landscape. Comparative studies across regions with diverse regulatory environments could offer deeper insights into how institutional quality and governance impact the financial inclusion-stability nexus. Additionally, future studies could employ mixed-method approaches, combining quantitative analysis with qualitative case studies, to capture the effects of socio-economic factors and consumer behavior on financial stability. Further exploration into the role of digital financial services and fintech innovations in shaping both financial inclusion and systemic risk would also be valuable, especially in the context of emerging technologies and cybersecurity challenges. Longitudinal studies examining post-crisis periods or major economic shocks could reveal how financial inclusion strategies perform under stress, offering critical guidance for resilient financial policymaking.

CRediT authorship contribution statement

Shaoming Han: Conceptualization, Methodology, Data curation, Writing – original draft. **Cheng Qian:** Supervision, Project administration, Writing – review & editing. **Nawal Abdalla Adam:** Formal analysis, Investigation, Visualization. **Norboy Karimov:** Validation, Resources, Data. **Wang Zhang:** Funding acquisition, Project administration, Supervision, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix B. Construction and Sources of Key Variables

Variable	Definition	Source
Z-score	Proxy for bank stability calculated as (ROA + Equity/Assets)/σ(ROA)	BankFocus/Orbis Bank database
FA_Index_A (FI_mm)	Composite financial inclusion index using mobile money and access indicators	IMF Financial Access Survey; World Bank
FA_Index_B (FI_sf)	Composite financial inclusion index using savings and usage indicators	Global Findex Database; World Bank
Operational Efficiency	Cost-to-income ratio and Net Interest Margin as indicators	BankFocus
Risk Exposure	Measured via the ratio of non-performing loans (NPL) to total loans	BankFocus/IMF Financial Soundness Indicators
Funding Stability	Core deposit to total liabilities ratio	BankFocus
ICT Access	Number of mobile/cellular subscriptions per 100 people	World Development Indicators (WDI)
Gender Equality	Percentage of female account holders or gender gap in financial access	Global Findex/World Bank Gender Data
Macro Controls	Includes GDP growth, inflation rate, and monetary policy rate	IMF WEO; World Bank
Bank Controls	Includes size (log of total assets), liquidity, capitalization, and efficiency	BankFocus
Rule of Law	Governance index capturing legal integrity and contract enforcement	World Governance Indicators (WGI)
Regulatory Quality	Measures the ability of the government to formulate sound banking policies	WGI
Deposit Insurance	Binary variable indicating existence of a national deposit insurance scheme	IADI; IMF
Bank Regulation	Quantitative score based on Basel compliance and supervision stringency	Barth et al. (Bank Regulation and Supervision Survey)
Income Groups	High-, middle-, and low-income country classifications	World Bank Country Classifications
Business Cycle	GDP cycle dummy (expansion/recession) and crisis period indicator	IMF/Laeven & Valencia Crisis Database

Data availability

Data will be made available on request.

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